Physics Worksheet

Momentum and Centre of Mass

Name: _____ **Student #:** _____ **Student Email #:** _____

1. Imagine a game of pool on a peculiar pool table. The white cue is struck and glides across a electromagnetic rail and strikes the side of the table where it compresses a spring, it then collides with the 8-ball and sends it into a pocket for the win. Below is a list of the forces acting on the **system of the two cue balls**. Explain whether these forces are external or internal. If they are external, explain what objects we would need to include in our system for them to be internal.

   (a) Applied force on the white cue from the player

   (b) Applied force of the collision between the two cue balls

   (c) The electromagnetic force from the rail

   (d) The spring froce

   (e) The gravity between the two cue balls

   (f) The friction between the table and the cue balls